

Inverse Transform Watermark

Remark 1.

Let X be a random variable with CDF $F_X(x) = \mathbb{P}(X \leq x)$.

Let $U \sim U(0, 1)$.

Let $Y = F_X^{-1}(U)$, where $F^{-1}(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}$, $\forall p \in [0, 1]$.

Then $Y \sim X$.

We see "unbiasedness" there.

How does it work?

Imposing an Order on the Vocabulary

The Inverse Transform method requires a 1D real number line to define a "Cumulative" function (i.e., you must be able to say $X \leq x$).

However, the vocabulary $\mathcal{W}$ is a set of words (categorical data) with no inherent mathematical order.

To solve this, we introduce a permutation $\pi : \mathcal{W} \rightarrow \{1, 2, \dots, |\mathcal{W}|\}$.

- $\pi(w)$ assigns a unique integer rank to a token w .
- $\pi^{-1}(i)$ takes rank i and returns the corresponding token. This permutation π allows us to arrange the discrete probabilities of the NTP distribution $\mathbf{P}$ into a 1D sequence on the integer line. The probability of the token at position i is $P_{\pi^{-1}(i)}$.

Remark 2.

Why need permutation?

Even if we use some fixed ordering, inverse transform sampling still works and the watermark is still unbiased (intuitively, only the "length" but not the position matters).

But we need this permutation to preserve security (if use some fixed order, then w_t and the hidden pseudorandom number U_t 's coupling relationship are static and clear.) and more importantly, validity of the pivotal quantity under H_0 .

Deriving the CDF $F(x; \pi)$

Consider the multinomial distribution with probability mass $P_{\pi^{-1}(i)}$ at i for $1 \leq i \leq |\mathcal{W}|$.

Now that the tokens are ordered from 1 to $|\mathcal{W}|$, we can define the CDF. The CDF $F(x; \pi)$ is the cumulative sum of probabilities for all tokens assigned a rank less than or equal to x .

For any real number x : $F(x; \pi) = \sum_{i=1}^{\lfloor x \rfloor} P_{\pi^{-1}(i)}$

To express this in terms of the original vocabulary $\mathcal{W}$ rather than the integer indices, we sum over all tokens $w' \in \mathcal{W}$, but use an indicator function $\mathbf{1}_{\{\pi(w') \leq x\}}$ to only include the probabilities of tokens whose rank is less than or equal to x .
 $F(x; \pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq x\}}$

Deriving the Generalized Inverse $F^{-1}(U; \pi)$

Because the distribution is discrete, the CDF $F(x; \pi)$ is a step function. we use the generalized inverse. For a uniform random variable $U \sim U(0, 1)$, $F^{-1}(U; \pi) = \min\{i \in \mathbb{Z} : F(i; \pi) \geq U\}$

Which is $F^{-1}(U; \pi) = \min\{i : \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq i\}} \geq U\}$

Deriving the Decoder $\mathcal{S}^{\text{inv}}$

$w_t = \mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) := \pi^{-1}(F^{-1}(U; \pi))$ where $\zeta = (\pi, U)$ and the permutation π is uniformly at random.

We show this watermark is unbiased.

We need to show $\mathbb{P}(\mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) = w) = P_w$ for all $w \in \mathcal{W}$.

Proof.

Let w be an arbitrary token in $\mathcal{W}$. Let its rank under the permutation be $k = \pi(w)$. Applying π to both sides of $\pi^{-1}(F^{-1}(U; \pi)) = \mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) = w$, equivalently $F^{-1}(U; \pi) = \pi(w) = k$.

And so equivalently $F(k-1; \pi) < U \leq F(k; \pi)$.

Since $F(k; \pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq k\}}$ and $F(k-1; \pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') < k\}}$, then equivalently

$$\sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') < k\}} < U \leq \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq k\}}$$

The length of this interval is $F(k; \pi) - F(k-1; \pi) = P_{\pi^{-1}(k)} = P_w$.

Since U is uniformly distributed on the interval $(0, 1)$, then $\mathbb{P}(\mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) = w) = \mathbb{P}(F(k-1; \pi) < U \leq F(k; \pi)) = P_w$.

Or we can use the first remark in this document.

$\mathbb{P}(X = i) = P_{\pi^{-1}(i)}$, if $Y = F_X^{-1}(U)$, then $Y \sim X$.

$\mathbb{P}(Y = i) = \mathbb{P}(X = i) = P_{\pi^{-1}(i)}$.

Note $\mathbb{P}(\text{output } w) = \mathbb{P}(Y = \pi(w))$, thus $\mathbb{P}(Y = \pi(w)) = \mathbb{P}(X = \pi(w)) = P_{\pi^{-1}(\pi(w))} = P_w$.

□

Main results

By "under H_1 , in contrast, a larger U_t value suggests a larger value of $\pi_t(w_t)$ in distribution for the watermarked text, which can be gleaned from (23)", it means that: recall from above, we know if say $\pi_t(w) = k$, then token w is chosen iff $F(k-1; \pi_t) < U_t \leq F(k; \pi_t)$.

Typo above equation (37). Should be $\pi(l) = w$ not $\pi(l) = k$ in the index of summation.

Proof of Lemma 4.1

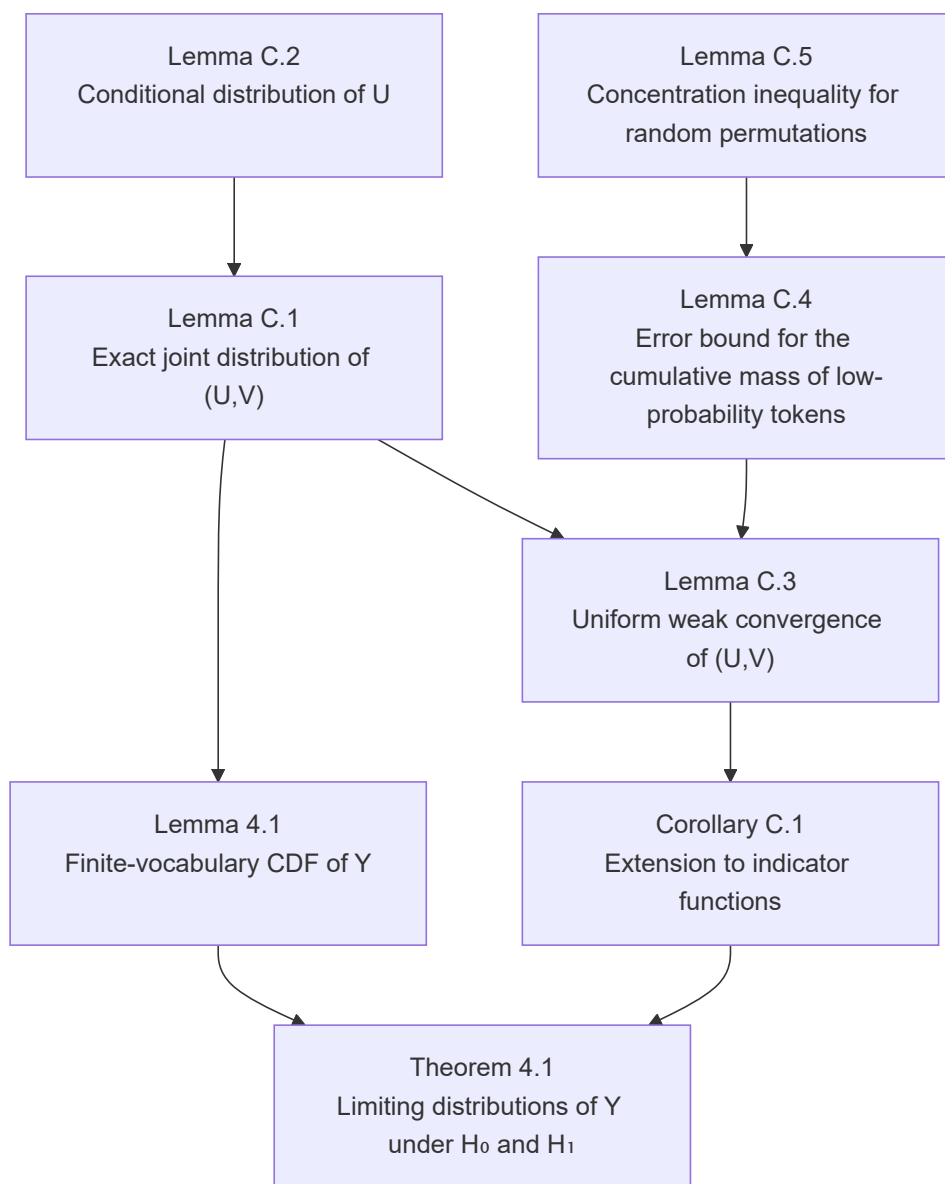
For H_0 , From

$$\mathbb{P}(Y_t^{\text{dif}} \leq r \mid H_0) = \sum_{w \in \mathcal{W}} \mathbb{P}(U_t \in [\eta(w) - r, \eta(w) + r] \cap [0, 1], \pi_t(w_t) = w \mid H_0)$$

Use results from lemma C.1 for H_0 : the joint probability is $\frac{1}{|\mathcal{W}|}$ multiplied by the probability of U_t falling into the target interval. Because $U_t \sim U(0, 1)$, the probability of U_t falling into a specific interval within $[0, 1]$ is exactly the length of that interval.

For H_1 , note changed order of summation and replaced probability to uniform measure.

Theorem 4.1



Now we first focus on Lemma C.5

Lemma C.5

First we look at Bernstein's Inequality.

Lemma 3 (Bernstein's Inequality).

Let $X_1, \dots, X_n$ be independent random variables satisfying

$$\mathbb{E}[X_i] = 0, \quad |X_i| \leq M \quad \text{almost surely}$$

for every i .

Define

$$S_n = \sum_{i=1}^n X_i, \quad V = \sum_{i=1}^n \mathbb{E}[X_i^2] = \sum_{i=1}^n \text{Var}(X_i).$$

Then, for every $t > 0$,

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + Mt/3)}\right)$$

Consequently,

$$\mathbb{P}(|S_n| \geq t) \leq 2 \exp \left(-\frac{t^2}{2(V + Mt/3)} \right)$$

Proof.

We will first show that, for every

$$0 < \lambda < \frac{3}{M}$$

we have

$$\mathbb{E}[e^{\lambda X_i}] \leq \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right)$$

Since X_i is bounded, by Fubini theorem, since

$$\sum_{k=0}^{\infty} \mathbb{E} \left[\left| \frac{\lambda^k X_i^k}{k!} \right| \right] \leq \sum_{k=0}^{\infty} \frac{(\lambda M)^k}{k!} = e^{\lambda M} < \infty.$$

Then

$$\mathbb{E} \left[\sum_{k=0}^{\infty} \frac{\lambda^k X_i^k}{k!} \right] = \sum_{k=0}^{\infty} \mathbb{E} \left[\frac{\lambda^k X_i^k}{k!} \right].$$

Therefore, by Taylor and above,

$$\begin{aligned} \mathbb{E}[e^{\lambda X_i}] &= \mathbb{E} \left[1 + \lambda X_i + \sum_{k=2}^{\infty} \frac{\lambda^k X_i^k}{k!} \right] \\ &= 1 + \lambda \mathbb{E}[X_i] + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}[X_i^k]}{k!}. \end{aligned}$$

Since $\mathbb{E}[X_i] = 0$,

$$\mathbb{E}[e^{\lambda X_i}] = 1 + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}[X_i^k]}{k!}.$$

For every integer $k \geq 2$,

$$\mathbb{E}[X_i^k] \leq \mathbb{E}[|X_i|^k].$$

Moreover, because $|X_i| \leq M$,

$$|X_i|^k = X_i^2 |X_i|^{k-2} \leq X_i^2 M^{k-2}$$

It follows that

$$\mathbb{E}[|X_i|^k] \leq M^{k-2} \mathbb{E}[X_i^2].$$

Thus,

$$\mathbb{E}[e^{\lambda X_i}] \leq 1 + \mathbb{E}[X_i^2] \sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{k!}$$

For every integer $k \geq 2$,

$$k! \geq 2 \cdot 3^{k-2}$$

This follows by induction. Equality holds for $k = 2$, and if it holds for k , then

$$(k+1)! = (k+1)k! \geq 3 \cdot 2 \cdot 3^{k-2} = 2 \cdot 3^{k-1}$$

Using $k! \geq 2 \cdot 3^{k-2}$, we obtain

$$\begin{aligned}\sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{k!} &\leq \sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{2 \cdot 3^{k-2}} \\ &= \frac{\lambda^2}{2} \sum_{k=2}^{\infty} \left(\frac{\lambda M}{3} \right)^{k-2}\end{aligned}$$

Since $\lambda M/3 < 1$, the geometric series converges and

$$\sum_{k=2}^{\infty} \left(\frac{\lambda M}{3} \right)^{k-2} = \frac{1}{1 - \lambda M/3}.$$

Therefore,

$$\sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{k!} \leq \frac{\lambda^2}{2(1 - \lambda M/3)}.$$

Substituting this into above gives

$$\mathbb{E}[e^{\lambda X_i}] \leq 1 + \frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)}.$$

Finally, using $1 + u \leq e^u$ for $u \geq 0$, we obtain

$$\mathbb{E}[e^{\lambda X_i}] \leq \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right).$$

Next, because $X_1, \dots, X_n$ are independent,

$$\begin{aligned}\mathbb{E}[e^{\lambda S_n}] &= \mathbb{E} \left[e^{\lambda \sum_{i=1}^n X_i} \right] \\ &= \mathbb{E} \left[\prod_{i=1}^n e^{\lambda X_i} \right] \\ &= \prod_{i=1}^n \mathbb{E}[e^{\lambda X_i}].\end{aligned}$$

Applying $\mathbb{E}[e^{\lambda X_i}] \leq \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right)$ to each factor gives

$$\begin{aligned}\mathbb{E}[e^{\lambda S_n}] &\leq \prod_{i=1}^n \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right) \\ &= \exp \left(\frac{\lambda^2 \sum_{i=1}^n \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right).\end{aligned}$$

Since

$$V = \sum_{i=1}^n \mathbb{E}[X_i^2],$$

we conclude that

$$\mathbb{E}[e^{\lambda S_n}] \leq \exp \left(\frac{\lambda^2 V}{2(1 - \lambda M/3)} \right)$$

for every $0 < \lambda < 3/M$.

Fix $t > 0$. For every $0 < \lambda < 3/M$, the monotonicity of the exponential function and Markov's inequality give

$$\begin{aligned}\mathbb{P}(S_n \geq t) &= \mathbb{P}(e^{\lambda S_n} \geq e^{\lambda t}) \\ &\leq e^{-\lambda t} \mathbb{E}[e^{\lambda S_n}].\end{aligned}$$

Using $\mathbb{E}[e^{\lambda S_n}] \leq \exp \left(\frac{\lambda^2 V}{2(1 - \lambda M/3)} \right)$,

$$\mathbb{P}(S_n \geq t) \leq \exp \left(-\lambda t + \frac{\lambda^2 V}{2(1 - \lambda M/3)} \right)$$

Assume first that $V > 0$, and choose

$$\lambda = \frac{t}{V + Mt/3}$$

This choice is admissible because

$$\frac{\lambda M}{3} = \frac{Mt/3}{V + Mt/3} < 1.$$

Let

$$D = V + \frac{Mt}{3}.$$

Then $\lambda = \frac{t}{D}$, and

$$1 - \frac{\lambda M}{3} = 1 - \frac{Mt/3}{D} = \frac{V}{D}.$$

Consequently,

$$\begin{aligned} -\lambda t + \frac{\lambda^2 V}{2(1 - \lambda M/3)} &= -\frac{t^2}{D} + \frac{(t^2/D^2)V}{2(V/D)} \\ &= -\frac{t^2}{D} + \frac{t^2}{2D} \\ &= -\frac{t^2}{2D}. \end{aligned}$$

Substituting $D = V + Mt/3$ into $\mathbb{P}(S_n \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2 V}{2(1 - \lambda M/3)}\right)$, we obtain

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + Mt/3)}\right).$$

If $V = 0$, then

$$\sum_{i=1}^n \mathbb{E}[X_i^2] = 0.$$

Since every term is nonnegative, $\mathbb{E}[X_i^2] = 0$ for every i , which implies $X_i = 0$ almost surely. Thus $S_n = 0$ almost surely, and the inequality is trivial.

This proves the one-sided Bernstein inequality.

The variables $-X_1, \dots, -X_n$ are also independent and satisfy

$$\mathbb{E}[-X_i] = 0, \quad | -X_i | \leq M, \quad \text{Var}(-X_i) = \text{Var}(X_i).$$

Applying the one-sided inequality to $-S_n$ gives

$$\mathbb{P}(S_n \leq -t) = \mathbb{P}(-S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + Mt/3)}\right).$$

Since event

$$\{|S_n| \geq t\} = \{S_n \geq t \sqcup S_n \leq -t\}$$

then

$$\begin{aligned} \mathbb{P}(|S_n| \geq t) &\leq \mathbb{P}(S_n \geq t) + \mathbb{P}(S_n \leq -t) \\ &\leq 2 \exp\left(-\frac{t^2}{2(V + Mt/3)}\right). \end{aligned}$$

This completes the proof. □

Remark 4.

We want to write the Bernstein inequality in the so called high-probability form.

$$\mathbb{P}(|S_n| \geq t) \leq 2 \exp \left(-\frac{t^2}{2(V + Mt/3)} \right) \stackrel{?}{\leq} 2e^{-x}$$

It is sufficient to require

$$\frac{t^2}{2(V + Mt/3)} \geq x$$

Since the denominator is positive,

$$t^2 \geq 2x \left(V + \frac{Mt}{3} \right)$$

Equivalently,

$$t^2 - \frac{2Mx}{3}t - 2Vx \geq 0$$

The positive root of the corresponding quadratic equation is

$$t_*(x) = \frac{Mx}{3} + \sqrt{2Vx + \frac{M^2x^2}{9}}$$

Thus, every $t \geq t_*(x)$ satisfies the inequality. This gives the exact form

$$\mathbb{P} \left(|S_n| \geq \frac{Mx}{3} + \sqrt{2Vx + \frac{M^2x^2}{9}} \right) \leq 2e^{-x}$$

Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, we have

$$\sqrt{2Vx + \frac{M^2x^2}{9}} \leq \sqrt{2Vx} + \frac{Mx}{3}$$

Therefore,

$$t_*(x) \leq \sqrt{2Vx} + \frac{2Mx}{3}$$

Choosing the slightly larger and simpler threshold

$$t = \sqrt{2Vx} + \frac{2Mx}{3}$$

gives

$$\mathbb{P} \left(|S_n| \geq \sqrt{2Vx} + \frac{2M}{3}x \right) \leq 2e^{-x}$$

Remark 5 (Lemma C.5 is the permuted version of Bernstein inequality).

Compare Bernstein to Lemma C.5,

$$\mathbb{P} \left(|S_n| \geq \sqrt{2Vx} + \frac{2M}{3}x \right) \leq 2e^{-x}$$

where $S_n = \sum_{i=1}^n X_i$.

$$\mathbb{P} \left(|Z - \mathbb{E}_\pi Z| \geq 2\sqrt{\left(\frac{1}{|\mathcal{W}|} \sum_{j,l=1}^{|\mathcal{W}|} b_{j,l}^2 \right)x} + \sup_{1 \leq j,l \leq |\mathcal{W}|} b_{j,l}x \right) \leq 8e^{1/16} \exp \left(-\frac{x}{16} \right)$$

where $Z = \sum_{j \in \mathcal{W}} b_{j,\bar{\pi}(j)}$, $\bar{\pi}$ a random uniform permutation in $\mathcal{W}$.

By random uniform permutation, it means that for every $\pi \in \Pi$,

$$\mathbb{P}(\bar{\pi} = \pi) = \frac{1}{|\mathcal{W}|!}$$

Consequently, for a fixed j , $\bar{\pi}(j)$ is uniform on $\{1, \dots, \mathcal{W}\}$, i.e.

$$\mathbb{P}(\bar{\pi}(j) = l) = \frac{1}{|\mathcal{W}|}$$

for $l = 1, \dots, |\mathcal{W}|$.

If we denote

$$V_{\text{perm}} = \frac{1}{|\mathcal{W}|} \sum_{j,l=1}^{|\mathcal{W}|} b_{j,l}^2 = \sum_{j=1}^n \mathbb{E}_{\bar{\pi}} \left[b_{j,\bar{\pi}(j)}^2 \right], \quad M_{\text{perm}} = \sup_{1 \leq j,l \leq |\mathcal{W}|} b_{j,l}$$

because (intuitively it is clear, as the permutation is uniform)

$$\begin{aligned} \mathbb{E}_{\bar{\pi}} \left[b_{j,\bar{\pi}(j)}^2 \right] &= \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \mathbb{P}(\bar{\pi}(j) = l) \\ &= \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \frac{1}{|\mathcal{W}|} \\ &= \frac{1}{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \end{aligned}$$

so

$$\begin{aligned} \sum_{j=1}^{|\mathcal{W}|} \mathbb{E}_{\bar{\pi}} \left[b_{j,\bar{\pi}(j)}^2 \right] &= \sum_{j=1}^{|\mathcal{W}|} \left(\frac{1}{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \right) \\ &= \frac{1}{|\mathcal{W}|} \sum_{j=1}^{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \\ &= \frac{1}{|\mathcal{W}|} \sum_{j,l=1}^{|\mathcal{W}|} b_{j,l}^2 \end{aligned}$$

We can see Bernstein and Lemma C.5 share the same form (if neglect constants), as

$$V = \sum_{i=1}^n \mathbb{E}[X_i^2], \quad |X_i| \leq M$$

One may ask, **why not just use Bernstein instead of Lemma C.5?**

This is because for Bernstein, it is required that $X_1, \dots, X_n$ to be independent, which is not the case in Lemma C.5.

Denote $Y_j = b_{j,\bar{\pi}(j)}$. They are not independent since for example once we know $\bar{\pi}(1) = l$, then for any $j \neq 1$, we cannot have $\bar{\pi}(j) = l$.

Lemma C.4

Lemma 6 (C.4).

Define

$$I_{\pi,w} = \mathbf{1}_{\pi^{-1}(1) > w} \quad \text{and} \quad A_{\bar{\pi},w} = \sum_{j=1}^w \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$$

Then

$$\max \left\{ \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|, \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}| \right\} \leq \sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right]$$

and

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

where $P_{(i)}$ is the i -th largest probability in P and c_0 is universal positive constant.

Proof.

The purpose of Lemma C.4 is to control the approximation error.

Note it should be

$$A_{\bar{\pi},w} = \sum_{j=1}^w \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$$

instead of $A_{\bar{\pi},w} = \sum_{j \in \mathcal{W}} \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$ in the original lemma.

It is worth noting that, the proof secretly assumed (after a fixed relabeling depending only on P)

$$P_1 = P_{(1)} = 1 - \Delta$$

This is WLOG and valid because any fixed relabeling depending only on P induces a bijection on the set of permutations, under which a uniform random permutation remains uniform. (The authors implicitly identify the vocabulary $\mathcal{W}$ with $[|\mathcal{W}|]$).

Note the definition of

$$\bar{\pi} \sim U(\Pi)$$

and $\bar{\pi}(j)$ denotes the token on the j -th entry of the permutation $\bar{\pi}$.

Thus,

$$I_{\bar{\pi},w} = \mathbf{1}_{\bar{\pi}^{-1}(1) > w} = \begin{cases} 1, & \text{if the maximum-probability token appears after position } w \text{ in the permutation } \bar{\pi} \\ 0, & \text{if the maximum-probability token appears in the first } w \text{ position in the permutation } \bar{\pi}. \end{cases}$$

and

$$A_{\bar{\pi},w} = \sum_{j=1}^w \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$$

is calculating: For a given realization of the random permutation $\bar{\pi}$, let $L = \bar{\pi}^{-1}(1)$ denote the position of the maximum-probability token, the difference between the probability mass actually accumulated up to position $\min\{w, L - 1\}$ and the amount that would have accumulated over the same positions if the remaining probability mass Δ were evenly divided among all the other tokens. Once the maximum-probability token appears, $A_{\bar{\pi},w}$ stops changing.

Now we look at what is

$$\max \left\{ \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|, \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}| \right\} \leq \sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right]$$

where $a_{\bar{\pi},w} = \sum_{j=1}^w P_{\bar{\pi}(j)}$.

We see that

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

is the expectation, with respect to the uniformly random permutation $\bar{\pi}$, of the maximum absolute difference, taken over all positions before the maximum-probability token appears, between the probability accumulated up to position w and what would have accumulated if the remaining probability mass Δ were evenly divided among all the other tokens.

Also note

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}|$$

To see it, let

$$L = \bar{\pi}^{-1}(1), \quad D_w = a_{\bar{\pi},w} - \frac{w\Delta}{N-1}.$$

Since $L = \bar{\pi}^{-1}(1)$ is a function of the random permutation $\bar{\pi}$, L **is itself a random variable**. Consequently, taking the expectation $\mathbb{E}_{\bar{\pi}}$ averages over the randomness of L **as well as** the remaining randomness in $\bar{\pi}$.

Then

$$I_{\bar{\pi},w} |D_w| = \begin{cases} |D_w|, & w < L, \\ 0, & w \geq L. \end{cases}$$

Thus

$$\sup_w I_{\bar{\pi},w} |D_w| = \sup_{w < L} |D_w|.$$

Whereas

$$A_{\bar{\pi},w} = \begin{cases} D_w, & w < L, \\ D_{L-1}, & w \geq L. \end{cases}$$

Thus

$$\sup_w |A_{\bar{\pi},w}| = \max \left\{ \sup_{w < L} |D_w|, |D_{L-1}| \right\} = \sup_{w < L} |D_w|.$$

Therefore

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}|$$

as desired.

Note

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \mathbb{E}_{\bar{\pi}} \sup_{w \leq L-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

Now we look at what is

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right]$$

The difference is that we fix the maximum-probability token at each position ℓ , and we take the sup over all ℓ in the end. Intuitively, we are taking the worst case over all possible positions ℓ . This is also why intuitively the inequality holds, since the LHS is averaging all positions.

Specifically:

Recall

$$\sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \sup_{w \leq L-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|.$$

Since $\bar{\pi} \sim U(\Pi)$, then

$$\mathbb{P}_{\bar{\pi}}(\bar{\pi}(\ell) = 1) = \frac{1}{|\mathcal{W}|}.$$

and so using tower rule,

$$\begin{aligned}
& \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi}, w} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \\
&= \mathbb{E}_{\bar{\pi}} \left[\mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi}, w} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| L \right] \right] \\
&= \sum_{\ell \in \mathcal{W}} \mathbb{P}_{\bar{\pi}}(L = \ell) \mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi}, w} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| L = \ell \right] \\
&= \sum_{\ell \in \mathcal{W}} \mathbb{P}_{\bar{\pi}}(\bar{\pi}(\ell) = 1) \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \\
&= \frac{1}{|\mathcal{W}|} \sum_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \\
&\leq \sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right].
\end{aligned}$$

The last inequality is due to

$$\frac{1}{|\mathcal{W}|} \sum_{\ell \in \mathcal{W}} x_{\ell} \leq \sup_{\ell \in \mathcal{W}} x_{\ell}$$

Finally we look at

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

We note that

$$\mathbb{E}_{\bar{\pi}} [a_{\bar{\pi}, w} | \bar{\pi}(\ell) = 1] = \frac{w\Delta}{|\mathcal{W}| - 1}$$

This is because:

Fix $w \leq \ell - 1$, for any $j \leq w$ and any $i \neq 1$, we have

$$\begin{aligned}
& \mathbb{P}_{\bar{\pi}} (\bar{\pi}(j) = i | \bar{\pi}(\ell) = 1) \\
&= \frac{\#\{\bar{\pi} : \bar{\pi}(\ell) = 1, \bar{\pi}(j) = i\}}{\#\{\bar{\pi} : \bar{\pi}(\ell) = 1\}} \\
&= \frac{(|\mathcal{W}| - 2)!}{(|\mathcal{W}| - 1)!} \\
&= \frac{1}{|\mathcal{W}| - 1}
\end{aligned}$$

Then

$$\begin{aligned}
\mathbb{E}_{\pi} [P_{\bar{\pi}(j)} | \bar{\pi}(\ell) = 1] &= \sum_{i=2}^{|\mathcal{W}|} P_i \mathbb{P}_{\pi} (\bar{\pi}(j) = i | \bar{\pi}(\ell) = 1) \\
&= \frac{1}{|\mathcal{W}| - 1} \sum_{i=2}^{|\mathcal{W}|} P_i \\
&= \frac{\Delta}{|\mathcal{W}| - 1}
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathbb{E}_{\bar{\pi}} [a_{\bar{\pi},w} | \bar{\pi}(\ell) = 1] &= \mathbb{E}_{\bar{\pi}} \left[\sum_{j=1}^w P_{\bar{\pi}(j)} \mid \bar{\pi}(\ell) = 1 \right] \\
&= \sum_{j=1}^w \mathbb{E}_{\bar{\pi}} [P_{\bar{\pi}(j)} | \bar{\pi}(\ell) = 1] \\
&= \sum_{j=1}^w \frac{\Delta}{|\mathcal{W}| - 1} \\
&= \frac{w\Delta}{|\mathcal{W}| - 1}
\end{aligned}$$

and we are done.

We observe that, conditional on $\bar{\pi}(\ell) = 1$, for each fixed $w \leq \ell - 1$,

$$|a_{\bar{\pi},w} - \mathbb{E}_{\bar{\pi}} [a_{\bar{\pi},w} | \bar{\pi}(\ell) = 1]| = \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

This form of "sum - its expectation" suggests that we need to use some sort of concentration inequality.

Now we need to find the "right" inequality to use.

We observe

$$0 \leq P_{\bar{\pi}(j)} \leq P_{(2)}$$

and

$$\frac{\Delta}{|\mathcal{W}| - 1} = \frac{1}{|\mathcal{W}| - 1} \sum_{i=2}^{|\mathcal{W}|} P_i$$

which implies

$$0 \leq \frac{\Delta}{|\mathcal{W}| - 1} \leq P_{(2)}$$

Combining above,

$$P_{\bar{\pi}(j)}, \frac{\Delta}{|\mathcal{W}| - 1} \in [0, P_{(2)}].$$

Since the difference between any two intergers in the interval $[0, P_{(2)}]$ is at most $P_{(2)}$,

$$\left| P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right| \leq P_{(2)}$$

This reminds us the bounded-summand in the Bernstein inequality (**One can also think of Hoeffding, but why the paper did not use Hoeffding?**).

But this raises another problem, $P_{\bar{\pi}(j)}$ is not independent.

The paper found Lemma C.5 to solve this problem.

To prove the last inequality in Lemma C.4:

As in the paper, let

$$b_{j,l} = P_l \mathbf{1}_{j \leq w, l \neq j}$$

Condition on $\bar{\pi}(\ell) = 1$ and under $w \leq \ell - 1$,

$$\begin{aligned}
Z &:= \sum_{j \in \mathcal{W}} b_{j, \bar{\pi}(j)} = \sum_{j \in \mathcal{W}} P_{\bar{\pi}(j)} \mathbf{1}_{j \leq w, \bar{\pi}(j) \neq 1} \\
&= \sum_{j=1}^w P_{\bar{\pi}(j)} \\
&= a_{\bar{\pi},w}
\end{aligned}$$

If $b_{j,l} \neq 0$, we must have $l \neq 1$. In this case, $P_l \leq P_{(2)}$, so

$$\sup_{j,l} b_{j,l} \leq P_{(2)}$$

We have

$$\begin{aligned} \sum_{j,l} b_{j,l}^2 &= \sum_{j,l} P_l^2 \mathbf{1}_{j \leq w, l \neq 1} \\ &= \sum_{j=1}^w \sum_{l \neq 1} P_l^2 \\ &= w \sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \end{aligned}$$

Condition on $\bar{\pi}(\ell) = 1$, apply Lemma C.5 to the remaining $|\mathcal{W}| - 1$ uniform permutation (because after conditioning on $\bar{\pi}(\ell) = 1$, $\bar{\pi}$ is no longer an uniform permutation on $\mathcal{W}$, so in this case cannot use Lemma C.5). Thus for any $x > 0$,

$$\mathbb{P}_{\bar{\pi}} \left(\left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq 2 \sqrt{\frac{w}{|\mathcal{W}| - 1} \left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) x} + P_{(2)} x \mid \bar{\pi}(\ell) = 1 \right) \leq 8e^{1/16} \exp \left(-\frac{x}{16} \right)$$

Given $0 < \delta < 1$, choose

$$x = 16 \log \left(\frac{8e^{1/16}}{\delta} \right)$$

And so

$$\delta = 8e^{1/16} \exp \left(-\frac{x}{16} \right)$$

in the above inequality.

Since

$$\frac{1}{|\mathcal{W}| - 1} \leq \frac{2}{|\mathcal{W}|}$$

We get

$$\mathbb{P}_{\bar{\pi}} \left(\left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\frac{w}{|\mathcal{W}|} \left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log \left(\frac{c_0}{\delta} \right)} + P_{(2)} c_0 \log \left(\frac{c_0}{\delta} \right) \mid \bar{\pi}(\ell) = 1 \right) \leq \delta$$

for a suitable c_0 .

Since

$$\frac{w}{|\mathcal{W}|} \leq 1$$

We get

$$\mathbb{P}_{\bar{\pi}} \left(\left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log \left(\frac{c_0}{\delta} \right)} + P_{(2)} c_0 \log \left(\frac{c_0}{\delta} \right) \mid \bar{\pi}(\ell) = 1 \right) \leq \delta$$

Note

$$\left\{ \sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq B \right\} = \bigcup_{w \leq \ell-1} \left\{ \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq B \right\},$$

where $B = \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log \left(\frac{c_0}{\delta} \right)} + P_{(2)} c_0 \log \left(\frac{c_0}{\delta} \right)$.

Note for a fixed ℓ , there exists at most $|\mathcal{W}|$ of w 's such that $w \leq \ell - 1$.

Since $\mathbb{P}(\bigcup_{i=1}^m E_i) \leq \sum_{i=1}^m \mathbb{P}(E_i)$ for event $E_1, \dots, E_m$,

$$\begin{aligned} & \mathbb{P}_{\bar{\pi}} \left(\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log\left(\frac{c_0}{\delta}\right) + P_{(2)} c_0 \log\left(\frac{c_0}{\delta}\right)} \middle| \bar{\pi}(\ell) = 1 \right) \\ & \leq \sum_{w \leq \ell-1} \delta \\ & = (\ell - 1)\delta \\ & \leq (|\mathcal{W}| - 1)\delta \\ & \leq |\mathcal{W}| \delta \end{aligned}$$

Now we choose

$$\delta = \frac{1}{|\mathcal{W}|^2}$$

then

$$|\mathcal{W}| \delta = \frac{1}{|\mathcal{W}|}$$

and

$$\log \frac{c_0}{\delta} = \log(c_0 |\mathcal{W}|^2)$$

We can write $c_0 \log(c_0 |\mathcal{W}|^2)$ as

$$c_0 \log(c_0 |\mathcal{W}|)$$

because c_0 is just a universal positive constant.

Therefore

$$\mathbb{P}_{\pi} \left(\sup_{w \leq \ell-1} \left| a_{\pi, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)} \middle| \bar{\pi}(\ell) = 1 \right) \leq \frac{1}{|\mathcal{W}|}$$

Note that condition on $\bar{\pi}(\ell) = 1$, for all $w \leq \ell - 1$,

$$0 \leq a_{\pi, w} \leq \Delta$$

and

$$0 \leq \frac{w\Delta}{|\mathcal{W}| - 1} \leq \Delta$$

Thus

$$0 \leq \sup_{w \leq \ell-1} \left| a_{\pi, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \leq \Delta \leq 1$$

which follows from a similar argument above.

Denote

$$X_{\ell} := \sup_{w \leq \ell-1} \left| a_{\pi, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

and

$$B := \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

Define

$$\mathcal{G}_{\ell} := \{X_{\ell} < B\}$$

and

$$\mathcal{G}_\ell^c = \{X_\ell \geq B\}$$

Then from before we have

$$\mathbb{P}_\pi(\mathcal{G}_\ell^c | \bar{\pi}(\ell) = 1) \leq \frac{1}{|\mathcal{W}|}$$

Note

$$\mathbb{E}_{\bar{\pi}}[X_\ell | \bar{\pi}(\ell) = 1] = \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell} | \bar{\pi}(\ell) = 1] + \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell^c} | \bar{\pi}(\ell) = 1]$$

On $\mathcal{G}_\ell$, $X_\ell < B$, so

$$X_\ell \mathbf{1}_{\mathcal{G}_\ell} \leq B \mathbf{1}_{\mathcal{G}_\ell}$$

Thus

$$\begin{aligned} \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell} | \bar{\pi}(\ell) = 1] &\leq B \mathbb{E}_{\bar{\pi}}[\mathbf{1}_{\mathcal{G}_\ell} | \bar{\pi}(\ell) = 1] \\ &= B \mathbb{P}_\pi(\mathcal{G}_\ell | \bar{\pi}(\ell) = 1) \\ &\leq B \end{aligned}$$

On $\mathcal{G}_\ell^c$, recall from above it is always true that $X_\ell \leq 1$, and so

$$X_\ell \mathbf{1}_{\mathcal{G}_\ell^c} \leq \mathbf{1}_{\mathcal{G}_\ell^c}$$

Thus

$$\begin{aligned} \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell^c} | \bar{\pi}(\ell) = 1] &\leq \mathbb{E}_{\bar{\pi}}[\mathbf{1}_{\mathcal{G}_\ell^c} | \bar{\pi}(\ell) = 1] \\ &= \mathbb{P}_\pi(\mathcal{G}_\ell^c | \bar{\pi}(\ell) = 1) \\ &\leq \frac{1}{|\mathcal{W}|} \end{aligned}$$

Combining above, we have

$$\mathbb{E}_{\bar{\pi}}[X_\ell | \bar{\pi}(\ell) = 1] \leq B + \frac{1}{|\mathcal{W}|}$$

which is exactly

$$\mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

The intuition here is that

$$0 \leq X \leq 1, \quad \mathbb{P}(X \geq B) \leq \varepsilon \quad \implies \quad \mathbb{E}[X] \leq B + \varepsilon$$

Since this holds for all fixed ℓ , and the RHS is free of ℓ , we have

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

And we are finally done.

□

Lemma C.3

Theorem 4.2

